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(54) **EXPANDABLE PANELBOARD BUS ASSEMBLIES AND METHODS OF ASSEMBLING SAME**

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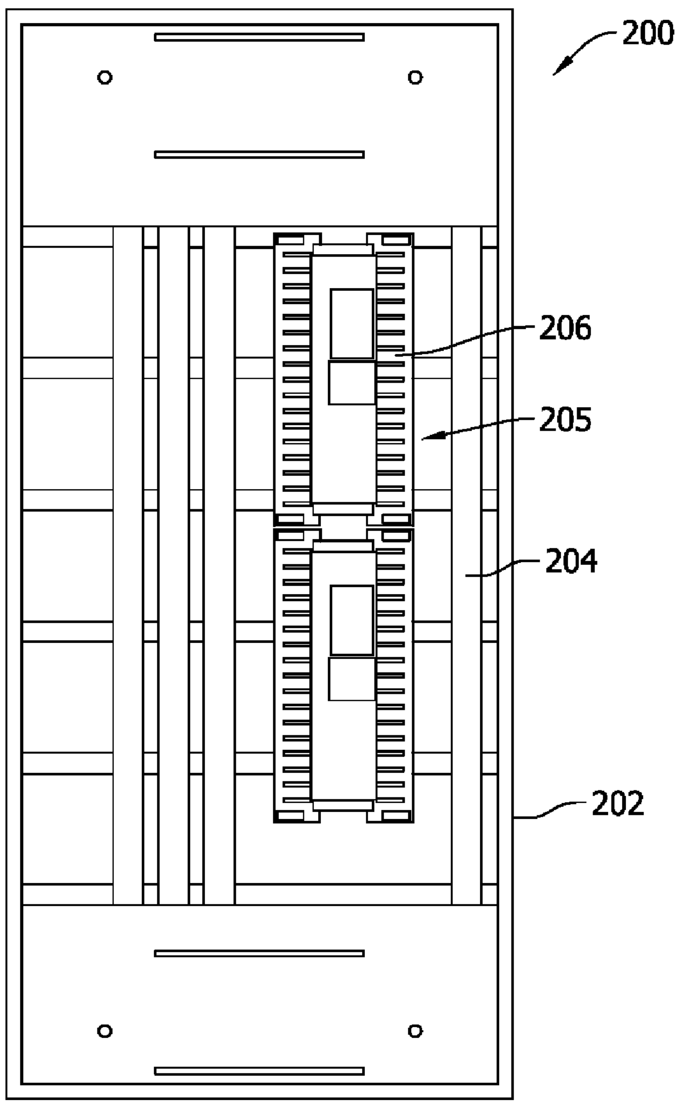
(57) **ABSTRACT**

An expandable electrical panelboard system is provided. The panelboard system includes a panelboard bus assembly. The panelboard bus assembly includes a bus subassembly including one or more busbar insulators and one or more panelboard busbars at least partially enclosed in the one or more busbar insulators. The panelboard bus assembly also includes an expansion subassembly including one or more expansion busbars. The one or more expansion busbars are electrically conductive and at least partially exposed. The bus subassembly is configured to mechanically and electrically couple with another bus subassembly via the expansion subassembly.



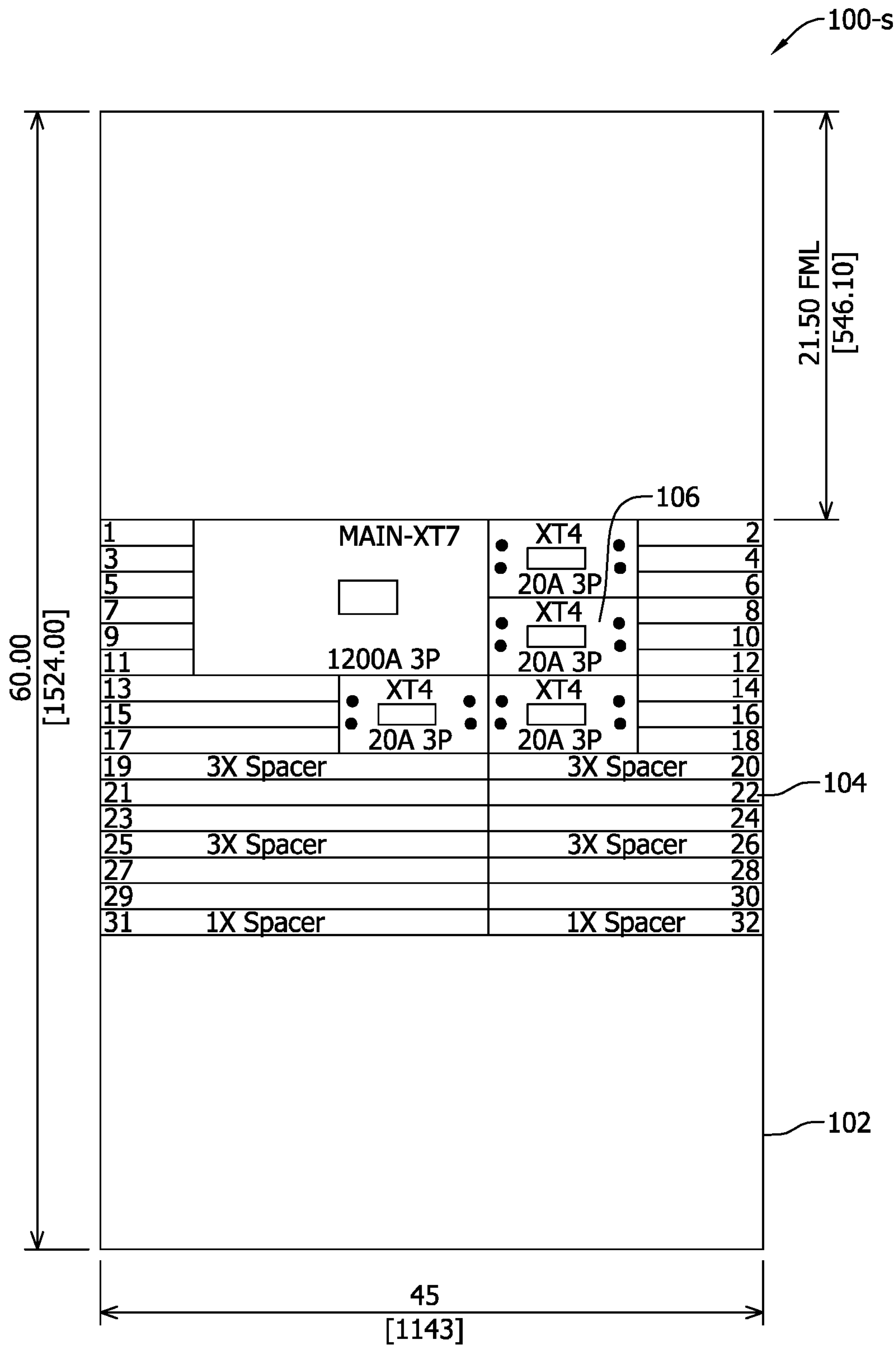


FIG. 1A
PRIOR ART

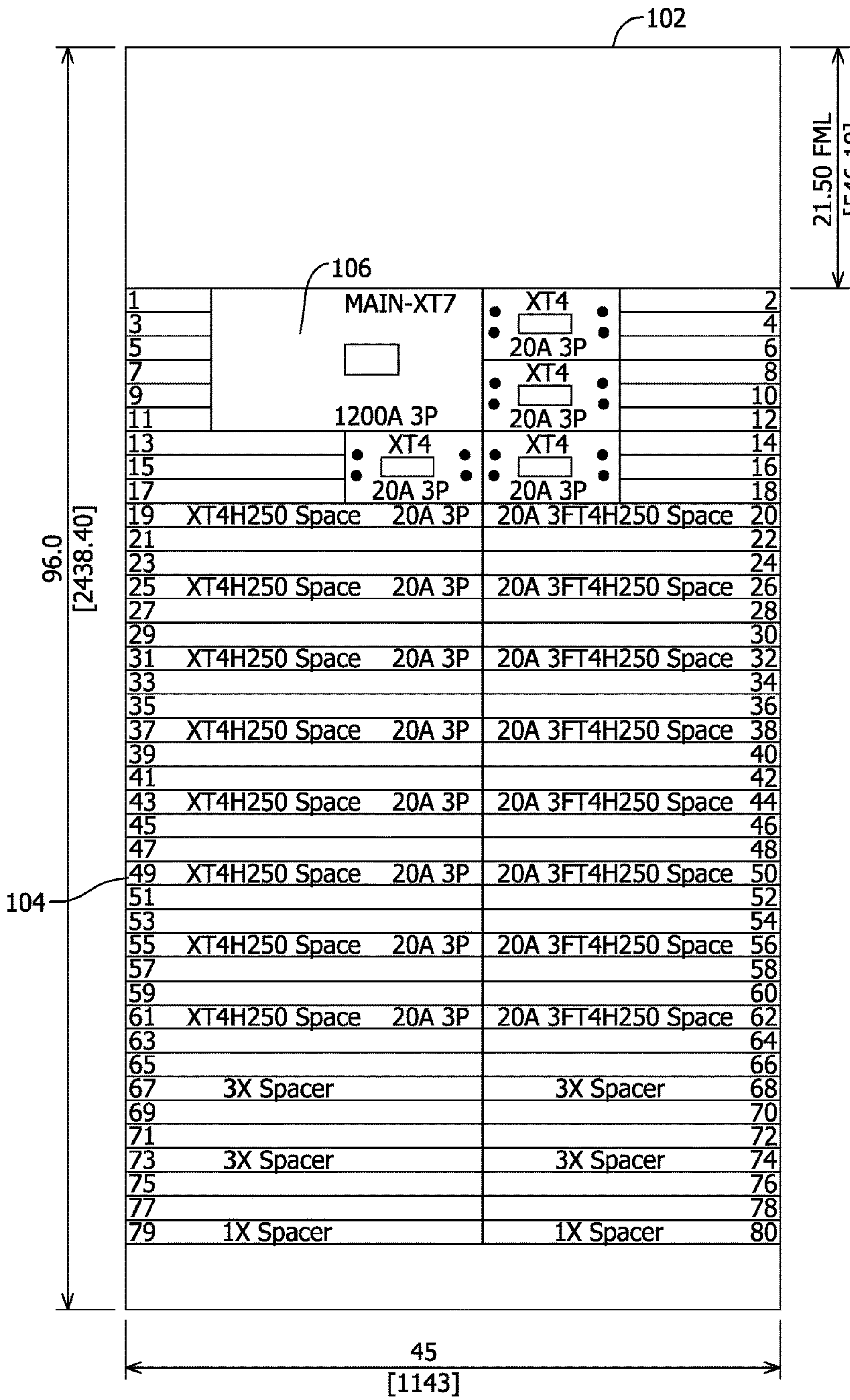


FIG. 1B
PRIOR ART

100-I

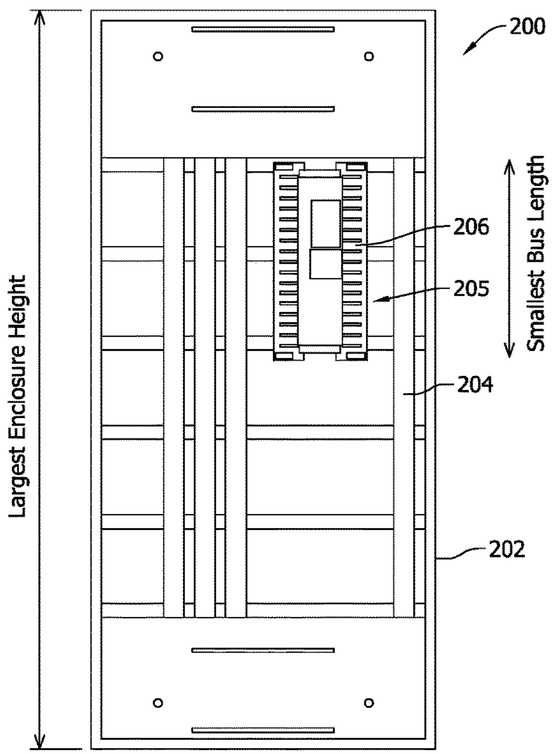

200
206
205
204
202
Smallest Bus Length
Largest Enclosure Height

FIG. 2A

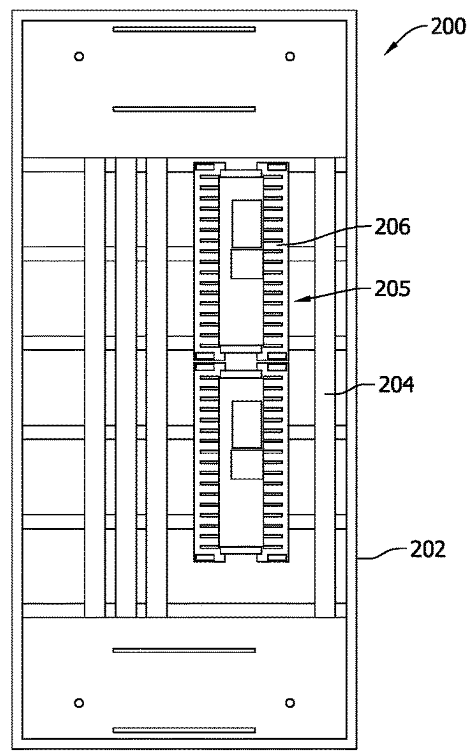

200
206
205
204
202

FIG. 2B

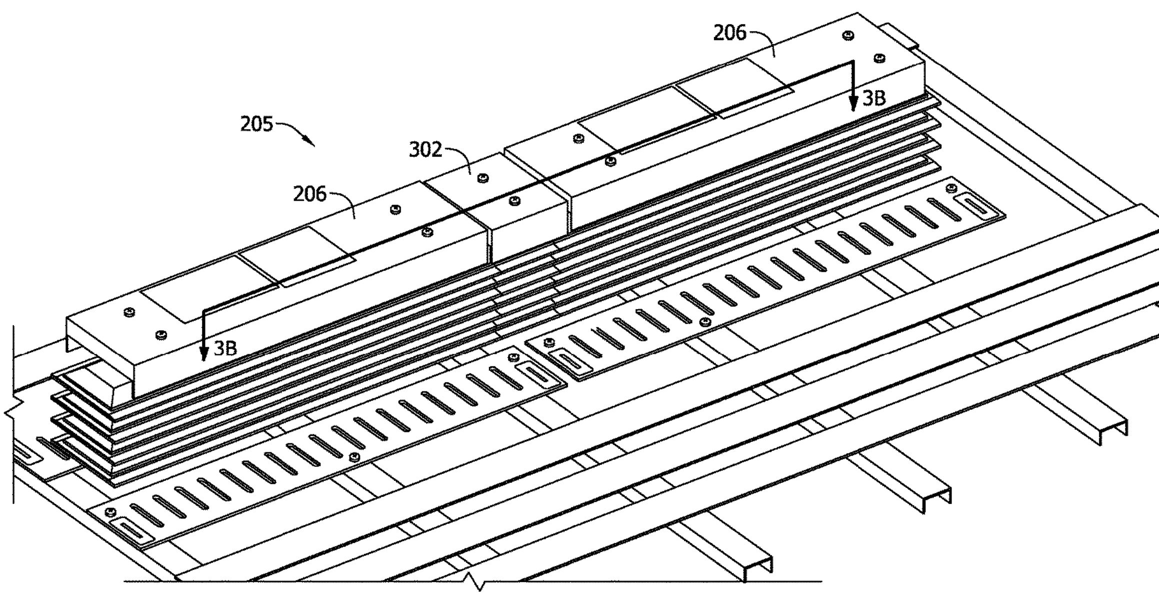


FIG. 3A

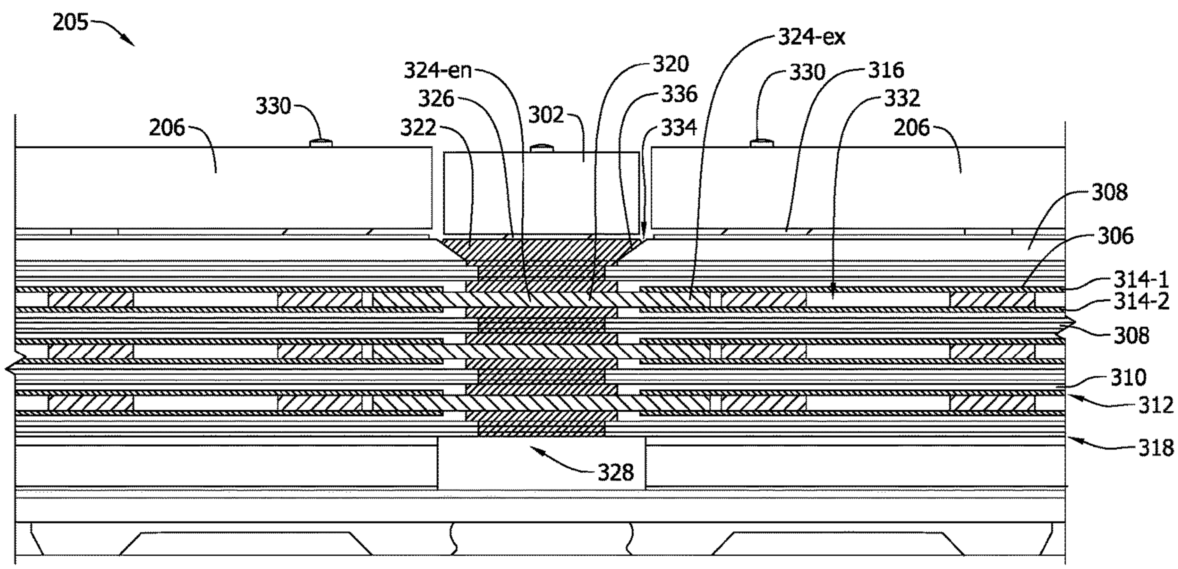

205
324-en
320
324-ex
330
316
326
336
332
302
334
322
206
308
306
314-1
314-2
310
312
318
328

FIG. 3B

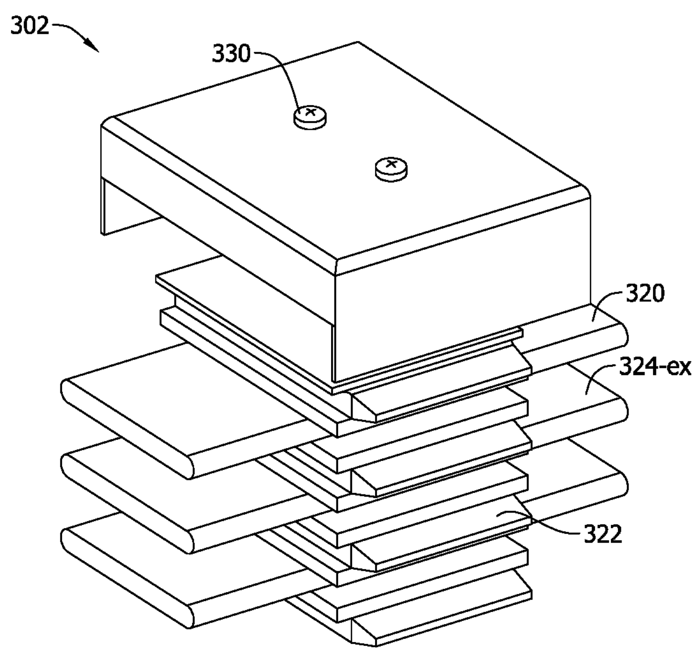


FIG. 3C

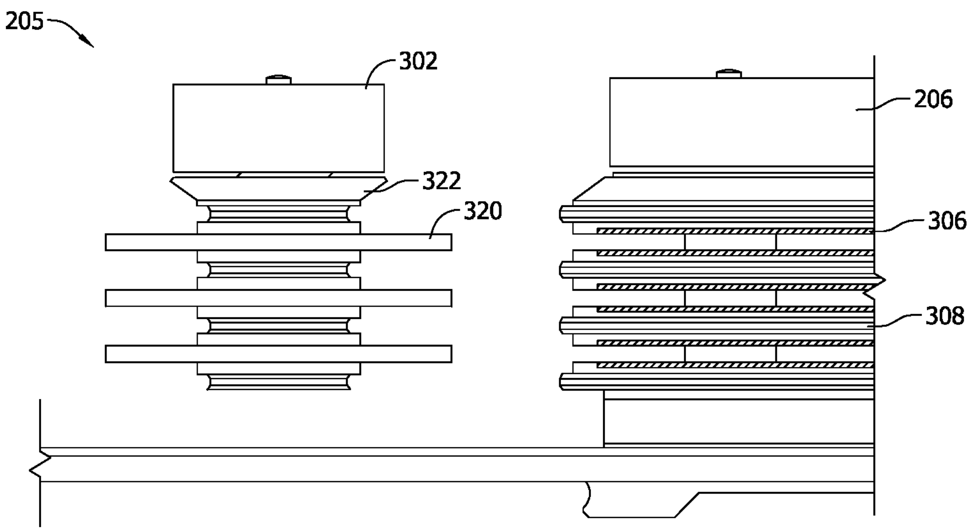


FIG. 3D

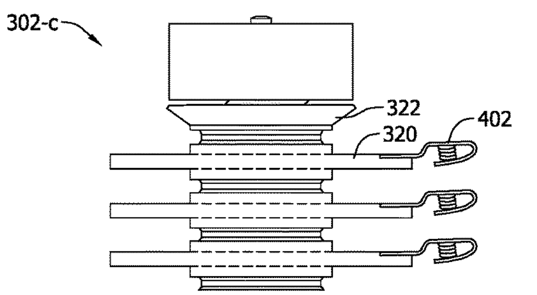


FIG. 4A

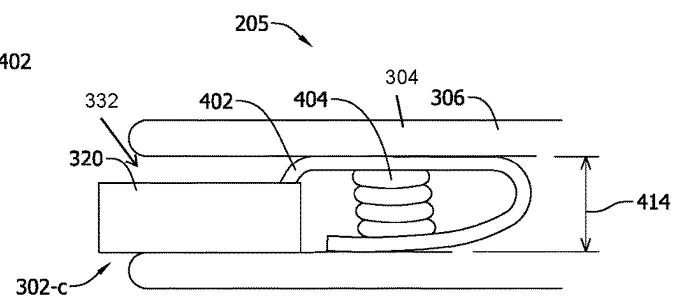


FIG. 4C

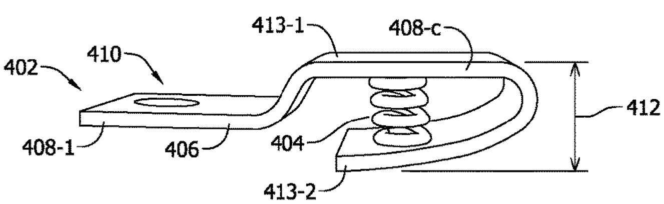


FIG. 4B

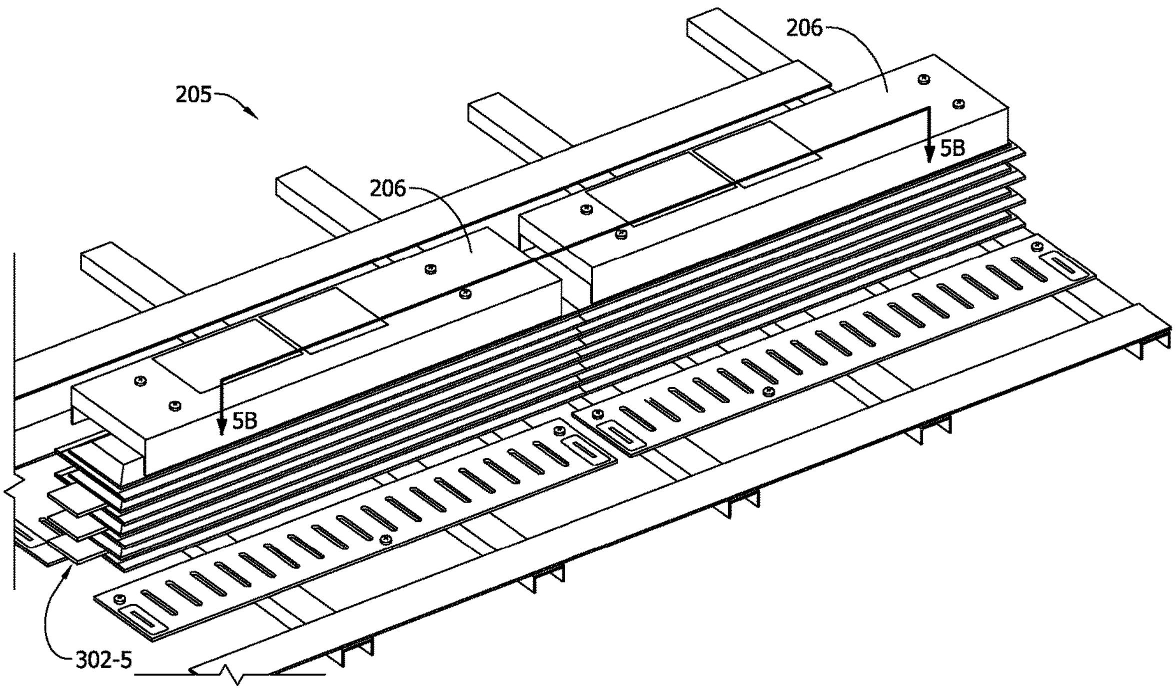


FIG. 5A

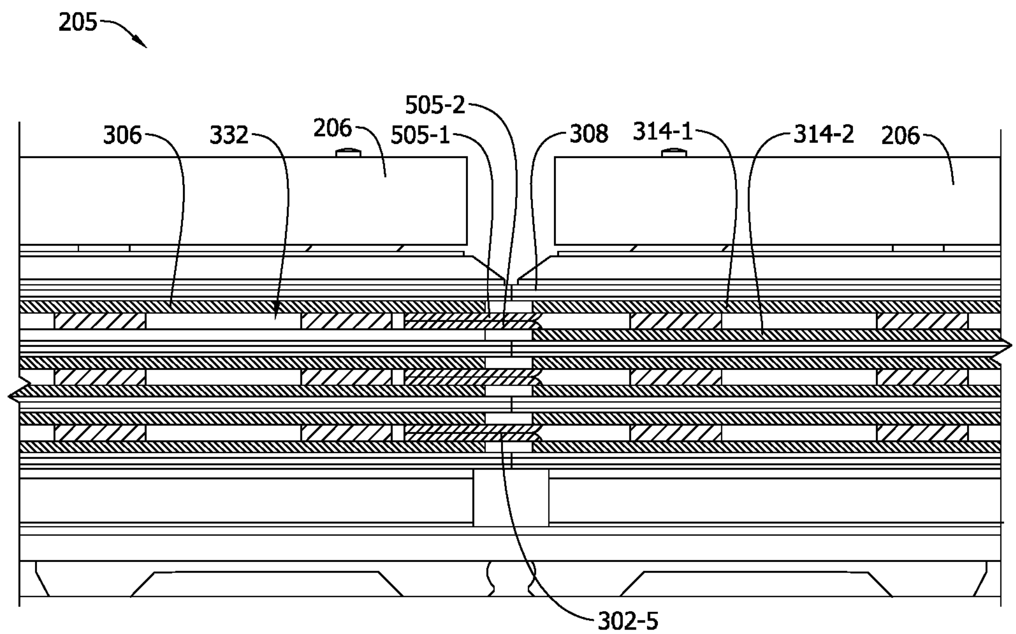


FIG. 5B

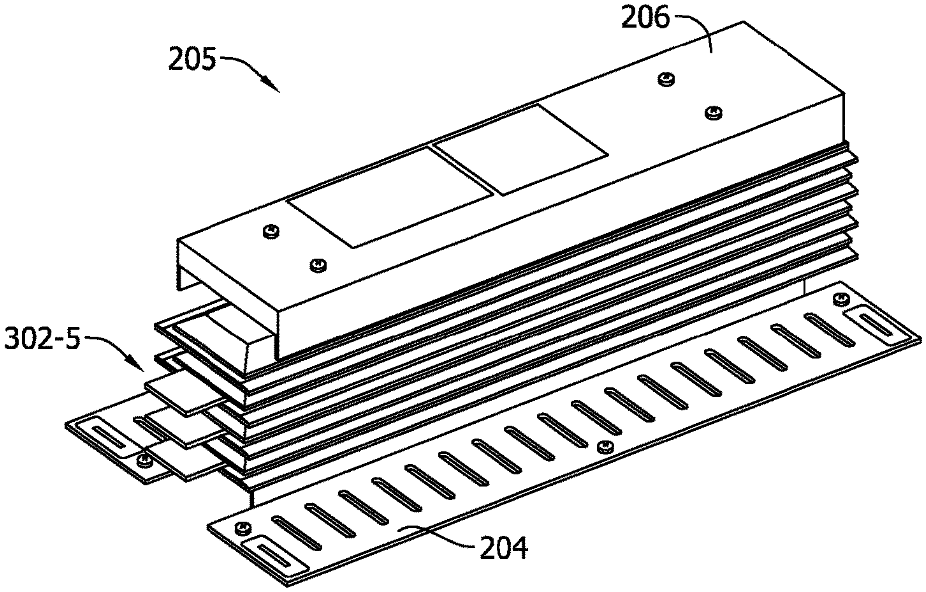

206
205
302-5
204

FIG. 5C

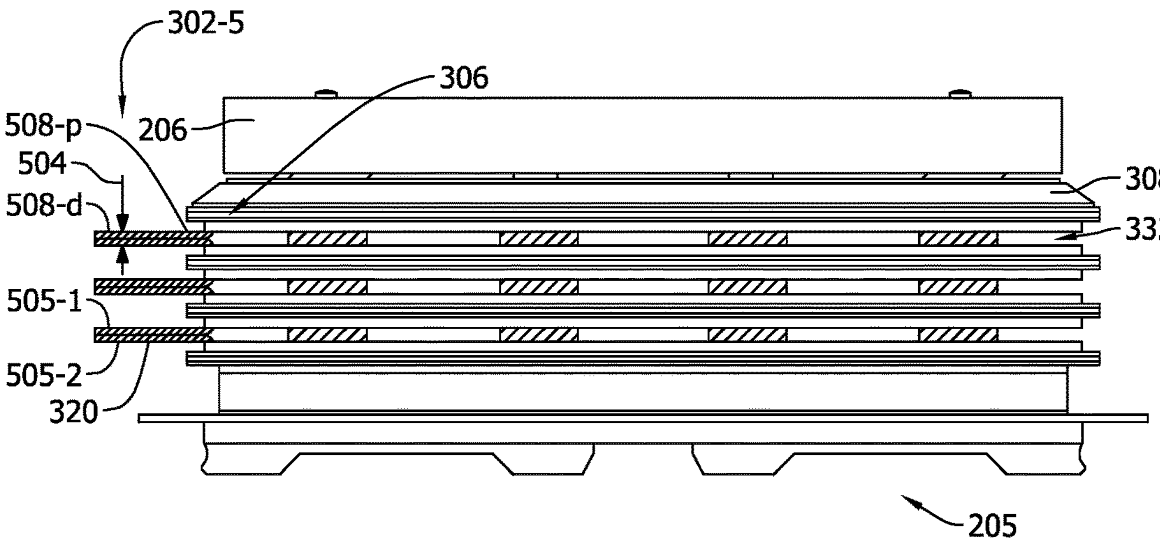

302-5
306
508-p
206
504
308
508-d
332
505-1
505-2
320
205

FIG. 5D

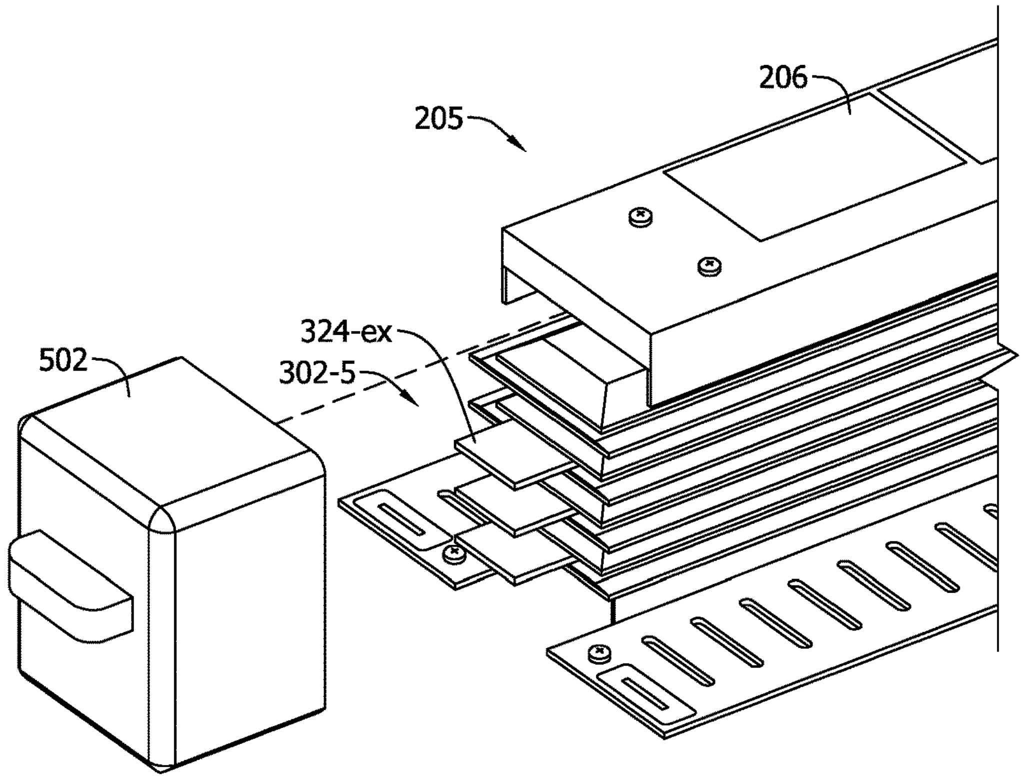


FIG. 5E

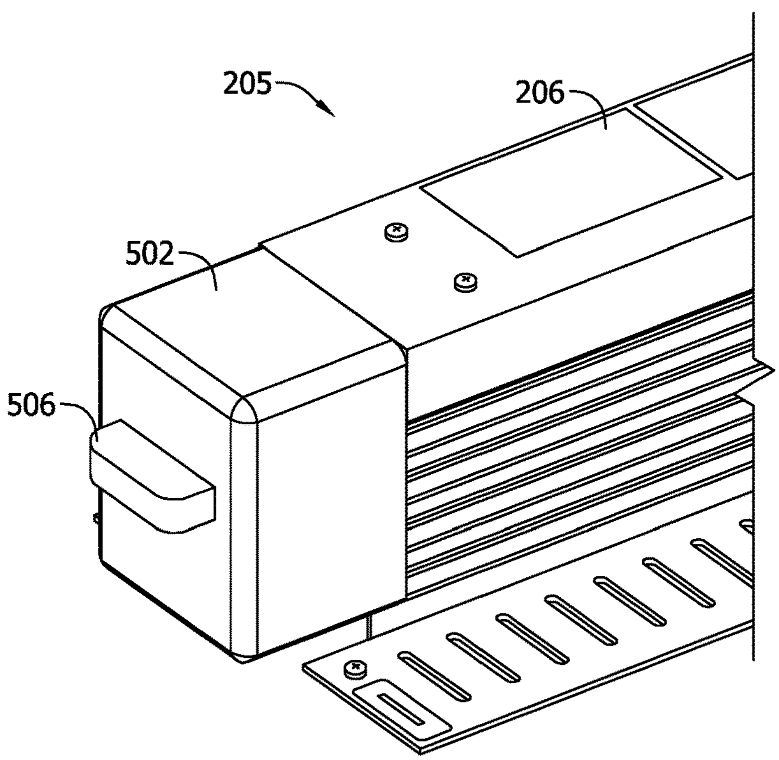


FIG. 5F

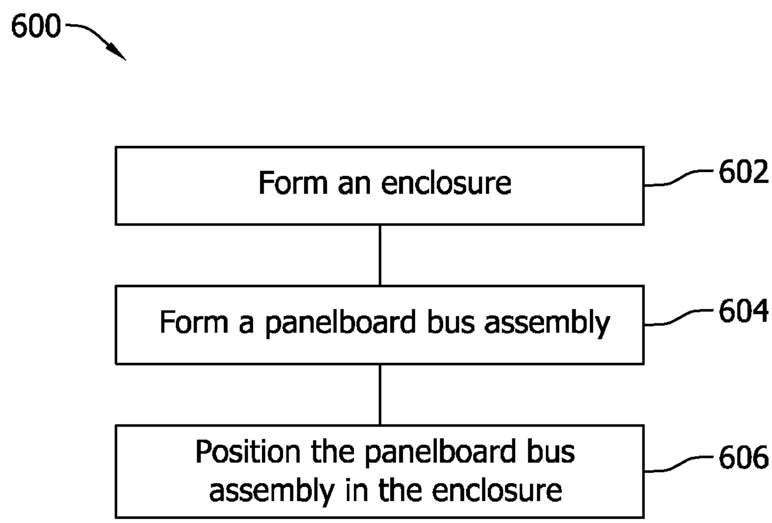

600
Form an enclosure
602
Form a panelboard bus assembly
604
Position the panelboard bus assembly in the enclosure
606

FIG. 6

EXPANDABLE PANELBOARD BUS ASSEMBLIES AND METHODS OF ASSEMBLING SAME

BACKGROUND

[0001] The field of the disclosure relates generally to electrical power delivery, and more particularly, to electrical panelboard systems, panelboard bus assemblies, and methods of assembling panelboard systems and assemblies.

[0002] A panelboard system in a power system is used to control the delivery of power from a power source to consumers. A panelboard system includes a panelboard bus assembly for transmitting electricity. Known panelboard systems are disadvantaged in some aspects and improvements are desired.

BRIEF DESCRIPTION

[0003] In one aspect, an expandable electrical panelboard system is provided. The panelboard system includes a panelboard bus assembly. The panelboard bus assembly includes a bus subassembly including one or more busbar insulators and one or more panelboard busbars at least partially enclosed in the one or more busbar insulators. The panelboard bus assembly also includes an expansion subassembly including one or more expansion busbars. The one or more expansion busbars are electrically conductive and at least partially exposed. The bus subassembly is configured to mechanically and electrically couple with another bus subassembly via the expansion subassembly.

[0004] In another aspect, an expandable panelboard bus assembly for an electrical panelboard system is provided. The panelboard bus assembly includes a bus subassembly including one or more busbar insulators and one or more panelboard busbars at least partially enclosed in the one or more busbar insulators. The panelboard bus assembly further includes an expansion subassembly including one or more expansion busbars, wherein the one or more expansion busbars are electrically conductive and at least partially exposed. The bus subassembly is configured to mechanically and electrically couple with another bus subassembly via the expansion subassembly.

[0005] In one more aspect, a method of assembling an expandable electrical panelboard system is provided. The method includes forming a panelboard bus assembly. The panelboard bus assembly includes a bus subassembly including one or more busbar insulators and one or more panelboard busbars at least partially enclosed in the one or more busbar insulators. The panelboard bus assembly further includes an expansion subassembly including one or more expansion busbars, wherein the one or more expansion busbars are electrically conductive and at least partially exposed. The bus subassembly is configured to mechanically and electrically couple with another bus subassembly via the expansion subassembly.

BRIEF DESCRIPTION OF DRAWINGS

[0006] These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings.

[0007] FIG. **1**A is a schematic diagram of a known panelboard system.

[0008] FIG. **1**B is a schematic diagram of another known panelboard system.

[0009] FIG. **2**A is a schematic diagram of an example panelboard system.

[0010] FIG. **2**B shows that the panelboard system illustrated in FIG. **2**A is expanded with another bus subassembly.

[0011] FIG. **3**A shows an example panelboard bus assembly of the panelboard system illustrated in FIG. **2**B.

[0012] FIG. **3**B is a cross-sectional view of the panelboard bus assembly shown in FIG. **3**A along cross-sectional line **3**B-**3**B as indicated in FIG. **3**A.

[0013] FIG. **3**C is a perspective view of an example expansion subassembly of the panelboard bus assembly shown in FIG. **3**A.

[0014] FIG. **3**D is a schematic diagram showing assembling of the panelboard bus assembly shown in FIG. **3**A by coupling of the expansion subassembly shown in

[0015] FIG. **3**C with an example bus subassembly.

[0016] FIG. **4**A is a schematic diagram of another example expansion subassembly.

[0017] FIG. **4**B shows an example clip of the expansion subassembly shown in FIG. **4**A.

[0018] FIG. **4**C is a schematic diagram showing coupling of the expansion subassembly shown in FIG. **4**A with an example bus subassembly.

[0019] FIG. **5**A shows one more example expandable panelboard bus assembly.

[0020] FIG. **5**B is a cross-sectional view of the panelboard bus assembly shown in FIG. **5**A along cross-sectional line **5**B-**5**B as indicated in FIG. **5**A.

[0021] FIG. **5**C shows one more example expansion subassembly for assembling the panelboard bus assembly shown in FIG. **5**A.

[0022] FIG. **5**D is a side view of the panelboard bus assembly shown in FIG. **5**C.

[0023] FIG. **5**E shows the expansion subassembly shown in FIG. **5**C to be coupled with an example insulation cap.

[0024] FIG. **5**F shows that the expansion subassembly illustrated in FIG. **5**C is coupled with the insulation cap.

[0025] FIG. **6** is a flow chart of an example method of assembling a panelboard systems shown in FIGS. **2**A-**5**F.

DETAILED DESCRIPTION

[0026] The disclosure includes expandable electrical panelboard systems, panelboard bus assemblies, and methods of assembling panelboard systems and panelboard bus assemblies. Method aspects will be in part apparent and in part explicitly discussed in the following description.

[0027] FIGS. **1**A and **1**B show known panelboard systems **100**-*s*, **100**-**1**. The known panelboard system **100** includes an enclosure **102** and a bus assembly **104**. The bus assembly **104** accommodates panelboard electronics **106**, such as breakers, which are for controlling delivery of electricity. Known panelboard systems **100** are not expandable. The user needs to choose either a small panelboard system **100**-*s* to minimize the costs, or a large panelboard system **100**-**1** for the future expansion. The costs of a panelboard system include the costs of the enclosure **102**, the bus assembly **104**, and panelboard electronics **106**. The costs for the sheet metal used in fabricating the enclosure **102** is in a relatively small percentage compared to the costs of the bus assembly **104**. Therefore, the costs of the large panelboard system **100**-**1** is much greater than the costs of the small panelboard system **100**-*s* although the panelboard systems **100** have the same setup of panelboard electronics **106** and the same capacity in power delivery. The value of the expandability in the large known panelboard system **100** is not realized until an uncertain future when the expansion actually occurs. On the other hand, when the need of expansion does arise, the user having the small panelboard system **100**-*s* needs to purchase and assemble a brand new panelboard system.

[0028] To expand the known panelboard system, significant costs from parts, labor, and downtime of the system are incurred. To install a larger panelboard system for accommodating more panelboard electronics, the user must power down the system and disconnect all cables and power connections. The old panelboard system needs to be removed before installing the larger panelboard system. Then, all the panelboard electronics and connections in the panelboard system are reinstalled and re-assembled. In an alternative traditional option, the user may purchase a new panelboard system and install the new panelboard system next to the existing panelboard system. The alternative option requires available space, which may be rare.

[0029] Accordingly, the nonexpendable known panelboard system **100** is costly in terms of the costs from parts, materials, labor, and/or downtime in addressing the need of expansion, which often arises as the consumers' need for electricity continues to grow.

[0030] The systems, assemblies, and methods described herein solve the above problems in at least some of the known panelboard systems. The panelboard systems and panelboard bus assemblies described herein are expandable, where bus subassemblies may be added or removed to accommodate the need without removing the old panelboard system, installing a brand new panelboard system, or leaving portions of the panelboard system unused after having incurred the upfront costs for a relatively large panelboard system. The existing components of the panelboard system, such as the enclosure, the framework, and the existing panelboard bus assemblies, may remain when the panelboard system is expanded, reducing costs in expanding a panelboard system in parts and labor. In expanding a panelboard bus assembly, a person in the field may face the challenge of ensuring the mechanical and electrical connections in the expanded panelboard bus assembly and the compliance with the standards requirements. The systems, assemblies, and methods described herein provide secure mechanical and electrical connections in the panelboard bus assemblies and comply with standard requirements governing panelboard systems and panelboard bus assemblies.

[0031] FIGS. **2**A and **2**B show an example panelboard system **200**. FIG. **2**A shows the panelboard system **200** at the initial setup. FIG. **2**B shows the panelboard system **200** when expanded. In the example embodiment, the panelboard system **200** includes an enclosure **202**. The enclosure **202** is fabricated with sheet metal such as aluminum or steel. The panelboard system **200** further includes a framework **204** distributed in the panelboard system **200**. The framework **204** provides structures onto which electronics and panelboard bus assemblies **205** are attached. The framework **204** defines various sizes and shapes of apertures and receptacles sized to receive fasteners, plugs, and/or connections for assembling the panelboard system **200**.

[0032] In the example embodiment, the panelboard system **200** further includes the panelboard bus assembly **205**. The panelboard bus assembly **205** includes a bus subassembly **206**. The panelboard bus assembly **205** is coupled with the framework **204**. At the initial phase, the system may only need one bus subassembly **206**. Later to accommodate a growing need, the panelboard bus assembly **205** is expanded with one or more bus subassemblies **206**. In the panelboard system **200**, the framework **204** and the enclosure **202** are constructed to include extra length in the framework **204** and extra space inside the enclosure **202** to facilitate the expansion.

[0033] In operation, to expand the panelboard system **200**, another bus subassembly **206** is included into the panelboard system **200** by attaching the new bus subassembly **206** with the existing panelboard bus assembly **205**. Electronics are assembled with the new bus subassembly **206**. The sizes of components in the panelboard system **200** may be standardized while still meeting changing needs of users. For example, only bus subassemblies **206** in a small number of sizes such as one or two are fabricated. The standardization greatly simplifies the manufacturing processes of the panelboard systems and panelboard bus assemblies, and the supply chain of the manufacturing facilities. The assembling of a panelboard system **200** is also simplified. Instead of removing an old panelboard system and assembling and installing a brand new panelboard system in at least some known systems and methods, during expansion, only additional bus subassemblies are assembled and installed, thereby reducing the costs in parts, material, and labor.

[0034] FIGS. **3**A-**3**D show example panelboard bus assembly **205** having example bus subassembly **206** and an example expansion subassembly **302**. FIG. **3**A shows an expanded panelboard bus assembly **205**. FIG. **3**B is a cross-sectional view of the panelboard bus assembly **205** shown in FIG. **3**A along cross-sectional line **3**B-**3**B as indicated in FIG. **3**A. FIG. **3**C shows the expansion subassembly **302**. FIG. **3**D is a schematic diagram of assembling the panelboard bus assembly **205** by coupling the expansion subassembly **302** with the bus subassembly **206**. Cross-sectional views of the expansion subassembly **302** and the bus subassembly **206** along cross-sectional line **3**B-**3**B are shown in FIG. **3**D.

[0035] In the example embodiments, the panelboard bus assembly **205** includes the bus subassembly **206** and the expansion subassembly **302**. The bus subassembly **206** includes one or more panelboard busbars **306** and one or more busbar insulators **308**. Each panelboard busbar **306** corresponds to a phase of the electricity. The panelboard busbar **306** is at least partially enclosed by the busbar insulator **308**. The busbar insulator **308** forms an elongated pocket **310** sized to receive the panelboard busbar **306** therein. The pocket **310** defines an insulator aperture **312** such that the panelboard busbar **306** at the insulator aperture **312** is exposed and may establish electrical connections with one or more electrically conductive components. The bus subassembly **206** complies with standards, such as Underwriters Laboratory (UL) 67, that govern electrical panelboards. For example, the panelboard busbars in the panelboard system are insulated by insulators such that the panelboard system meets a finger safe rating and prevents access to energized parts, thereby ensuring safety of the panelboard system and the field workers. In some embodiments, for one phase of electricity, the panelboard busbar **306** are split into two panelboard sub-busbars **314**, where a first panelboard sub-busbar **314-1** is at least partially enclosed in one busbar insulator **308** and a second panelboard sub-busbar **314-2** is at least partially enclosed in another busbar insulator **308**. A panelboard busbar **306** being split into two panelboard sub-busbars **314** is advantageous in achieving a desired cross-sectional area of the conductor of the panelboard busbar **306** for carrying current at a certain rating while reducing one or more dimensions of the conductor. The busbar insulators **308** and the panelboard busbars **306** may form into a busbar stack **318**, where one busbar insulator **308** is positioned over another busbar insulator **308** and one panelboard busbar **306** positioned over another panelboard busbar **306** in the radial direction.

[0036] In the example embodiments, the panelboard system **200** further includes a busbar bolt **316**. The busbar bolt **316** is coupled with the busbar stack **318** of busbar insulators **308** and panelboard busbars **306**. The busbar bolt **316** may be a fastener and extend through the busbar stack **318**. The torque of the busbar bolt **316** for compressing the busbar stack **318** may be adjusted by fastening or loosening the busbar bolt **316** in a desired amount. The busbar bolt **316** may be threaded. The torque may be adjusted by rotating the busbar bolt **316** into or out of the busbar stack **318** in a desired amount. Alternatively or additionally, the busbar insulators **308** and the panelboard busbars **306** are assembled together via other mechanism such as using adhesives.

[0037] In the example embodiment, the expansion subassembly **302** includes one or more expansion busbars **320**. Each expansion busbar may correspond to a phase of the electricity. The expansion busbars **320** are electrically conductive. The expansion busbars **320** are at least partially exposed, where at least part of the expansion busbars **320** are not insulated before the expansion subassembly **302** is coupled with other components of the panelboard system **200**. As used herein, an electrically conductive component is exposed when the component is not covered with an insulative material before the component is coupled with another component. The expansion subassembly **302** further includes one or more expansion insulators **322**. Each expansion busbar **320** is at least partially enclosed by the one or more expansion insulators **322**, where at least part of the expansion busbar **320** is surrounded and covered by the expansion insulators **322**. As a result, the expansion busbar **320** includes an exposed portion **324**-*ex* and an enclosed portion **324**-*en*, where the exposed portion **324**-*ex* is exposed from the expansion insulators **322** and the enclosed portion **324**-*en* is enclosed in the expansion insulators **322** and insulated by the expansion insulators **322**. The expansion busbar **320** includes two exposed portions **324**-*ex* with the enclosed portion **324**-*en* positioned therebetween, facilitating expansion of the panelboard bus assembly **205**, where two bus subassemblies **206** may be coupled with the expansion subassembly **302**. In the depicted embodiment, the expansion insulators **322** have a side shape or the shape of a side complementary with the side shape of the bus subassembly **206** (see FIG. 3B). An expansion stack **328** may be formed as one expansion insulator **322** is positioned over another expansion insulator **322** and one expansion busbar **320** is positioned over another expansion busbar **320** in the radial direction.

[0038] In the example embodiment, the expansion subassembly **302** further includes an expansion bolt **326**. The expansion bolt **326** is coupled with the expansion stack **328** of expansion insulators **322** and expansion busbars **320**. For example, the expansion bolt **326** may be a fastener and extend through the expansion stack **328**. The torque of the expansion bolt **326** for compressing the expansion stack **328** may be adjusted by fastening or loosening the expansion bolt **326** in a desired amount. The expansion bolt **326** may be threaded. The torque may be adjusted by rotating the expansion bolt **326** into or out of the expansion stack **328** in a desired amount. Alternatively or additionally, the expansion insulators **322** and the expansion busbars **320** are assembled together via other mechanism such as using adhesives.

[0039] In the example embodiment, the expansion subassembly **302** further includes an indicator **330** used to indicate whether the expansion subassembly **302** is energized. For example, if electricity passes through the expansion subassembly **302**, the indicator **330** is lit, and when no electricity passes through the expansion subassembly **302**, the indicator **330** is not lit. The indicator **330** may serve as a warning to a field worker about energized busbars. Indicators **330** may be included in bus subassemblies **206** for the warning purposes.

[0040] In operation, to couple the expansion subassembly **302** with the bus subassembly **206**, the expansion busbars **320** are aligned with the gaps **332** defined between the two panelboard sub-busbars **314** of a panelboard busbar **306**. The expansion busbars **320** are inserted into the gaps **332** with one expansion busbar **320** being inserted into one gap **332**. The expansion subassembly **302** may be inserted into the bus subassembly **206** as far as the gaps **332** and the expansion busbars **320** permit. In the depicted embodiment, the expansion subassembly **302** has a side shape complementary with the side shape of the bus subassembly **206** such that when the expansion subassembly **302** is assembled with the bus subassembly **206**, the spacing **334** between the expansion subassembly **302** and the bus subassembly **206** is relatively small. The side shapes of the expansion subassembly **302** and the bus subassembly **206** may be irregular such as being zigzagged or serpentine, thereby reducing or restricting the entry of substance such as water into the panelboard bus assembly **205**. The expansion subassembly **302** and the bus subassembly **206** are interwoven with one another, where some part of the bus subassembly **206** is inserted into the expansion subassembly **302** and some part of the expansion subassembly **302** is inserted into the bus subassembly **206**. In some embodiments, the coupling points **336** between the expansion subassembly **302** and the bus subassembly **206** may be sealed with a sealant, further limiting the entry of substance into the panelboard bus assembly **205**. Another bus subassembly **206** may be coupled with the expansion subassembly **302** at the other side of the expansion subassembly **302**, thereby expanding the panelboard bus assembly **205**.

[0041] When assembled, the expansion subassembly **302** and the bus subassembly **206** are electrically coupled with one another, where the expansion busbars **320** are coupled with the panelboard busbars **306**. For example, the expansion busbars **320** are inserted into the gap **332** between two panelboard sub-busbars **314** of a panelboard busbar **306**, establishing electrical connection. The mechanical and electrical connections and coupling between the expansion subassembly **302** and the bus subassembly **206** are strengthened by the busbar bolt **316** and/or the expansion bolt **326**. The coupling strength may be adjusted by adjusting the busbar bolt **316** and/or the expansion bolt **326**. Further, the compliance with the standard requirements governing the panelboard systems is maintained with panelboard systems **200**

and panelboard bus assemblies **205**. When the expansion subassembly **302** and the bus subassembly **206** are assembled together, exposed portions **324**-*ex* of the expansion subassembly **302** are insulated by the busbar insulators **308** of the bus subassembly **206**. The expanded panelboard bus assemblies **205** are finger safe, providing protection to the field worker from accidentally touching of energized panelboard systems and panelboard bus assemblies. Expanded panelboard bus assemblies **205** along with expansion subassemblies **302** may undergo temperature rise testing of the bus structures to ensure even temperature distribution and meets the requirements on thermal limits under the standards such that heat from the panelboard bus assemblies **205** and expansion subassemblies **302** is effectively dissipated.

[0042] FIGS. **4**A-**4**C show that the coupling between the expansion subassembly **302** and the bus subassembly **206** may be strengthened via a clip **402**. FIG. **4**A shows another example expansion subassembly **302**-*c*. FIG. **4**B shows the example clip **402**. FIG. **4**C shows that an expansion busbar **320** is coupled with a panelboard busbar **306** of a bus subassembly **206** using the clip **402**. In the example embodiment, different from the expansion subassembly **302** shown in FIGS. **3**A-**3**D, the expansion subassembly **302**-*c* includes the clip **402**. The clip **402** is fabricated with electrically conductive material such as copper. The clip **402** may include a spring **404**. The clip **402** further includes a clip body **406**. The clip body **406** may extend from one clip end **408-1** to the other clip end **408**-*c*. The clip body **406** may curve into a curved clip end **408**-*c* and form into a first clip portion **413-1** and a second clip portion **413-2**. In some embodiments, the first clip end **408-1** branches into a fork having the first clip portion **413-1** and the second clip portion **413**. The spring **404** is attached between the first clip portion **413-1** and the second clip portion **413-2**. In other embodiments, the clip **402** does not include a spring **404**, where the clip body **406** is fabricated with material having increased elasticity and functions as a spring. For example, the first clip portion **413-1** and the second clip portion **413-2** are compressed toward one another when the clip **402** is being inserted into the gap **332** between panelboard sub-busbars **304**, and a force is generated from the compression.

[0043] In operation, one or more clips **402** are coupled with the expansion busbar **320**. A clip aperture **410** may be defined at the first clip end **408** of the clip **402**. An expansion aperture (not show) may be defined in the expansion busbar **320**. The clip **402** may be coupled with the expansion busbar **320** by fastening a fastener into the apertures. The clip **402** may be coupled with the expansion busbar **320** via other mechanism such as soldering.

[0044] When the expansion subassembly **302** is coupled with the bus subassembly **206**, the clip **402** is received in the gap **332** between two panelboard sub-busbars **314** of the panelboard busbar **306**. The distance **412** between the first clip portion **413-1** and the second clip portion **413-2** when the spring is not compressed is greater than the width **414** of the gap **332**. As a result, when the clip **402** is received in the gap **332**, the spring **404** and/or the clip body **406** are compressed, and the force from the compressed spring **404** and/or the compressed clip body **406** presses the clip **402** against the panelboard busbar **306**, ensuring the electrical connection between the expansion subassembly **302** and the bus subassembly **206**. Using clips **402** to establish electrical and mechanical connection between the expansion subassembly **302** and the bus subassembly **206** may relax the tolerance requirements on the thickness of the expansion busbar **320** and the gap **332** in the bus subassembly **206** (see FIG. **4**C).

[0045] FIGS. **5**A-**5**F show one more example panelboard bus assembly **205** that includes the bus subassembly **206** and one more example expansion subassembly **302-5**. FIG. **5**A shows that a bus subassembly **206** is coupled with another bus subassembly **206**. FIG. **5**B is a cross-sectional view of a portion of the assembly shown in FIG. **5**A along cross-sectional line **5**B-**5**B as indicated in FIG. **5**A. FIG. **5**C shows the bus subassembly **206** and the expansion subassembly **302-5** coupled with a framework **204**. FIG. **5**D is a side view of the assembly shown in FIG. **5**C. FIG. **5**E shows the assembly illustrated in FIG. **5**C to be coupled with an example insulation cap **502**. FIG. **5**F shows the insulation cap **502** being coupled with the assembly illustrated in FIG. **5**C.

[0046] In the example embodiment, compared to expansion subassemblies **302** shown in FIGS. **3**A-**4**C, where the expansion subassembly **302** is a distinct component from the bus subassembly **206**, the expansion subassembly **302-5** is formed into one unit with the bus subassembly **206**. The expansion subassembly **302-5** includes expansion busbars **320**. The expansion busbars **320** extend from the panelboard busbars **306**. Each expansion busbar **320** may include a first expansion sub-busbar **505-1** and a second expansion sub-busbar **505-2**. The first expansion sub-busbar **505-1** extends from the first panelboard sub-busbar **314-1** (see FIG. **5**B). The second expansion sub-busbar **505-2** extends from the second panelboard sub-busbar **314-2**. The expansion busbars **320** are at least partially exposed, where the exposed portion **324**-*ex* (see FIG. **5**E) of the expansion busbar **320** is not insulated, when the expansion subassembly **302-5** is not coupled with another component of the panelboard system **200**, such as another bus subassembly **206**.

[0047] In operation, to expand the panelboard bus assembly **205**, the panelboard bus assembly **205** may be coupled with another bus subassembly **206** or another panelboard bus assembly **205** with the bus subassembly **206** and the expansion subassembly **302-5** being formed as one unit. To couple the two panelboard bus assemblies **205** or the panelboard bus assembly **205** with another bus subassembly **206**, the expansion busbars **320** of the expansion subassembly **302** are inserted into gaps **332** between neighboring busbar insulators **308**. The expansion busbars **320** electrically and mechanically couple with the panelboard busbars **306** in the other bus subassembly **206**. In some embodiments, the thickness **504** of the expansion busbar **320** at a distal end **508**-*d* of the expansion busbar **320** distal to the panelboard busbar **306** is greater than the thickness **504** of the expansion busbar **320** at a proximal end **508**-*p* of the expansion busbar **320** proximal to the panelboard busbar **306** (see FIG. **5**D). The increased thickness **504** at the distal end **508** may be formed by bending the two expansion sub-busbars **505** away from one another at the distal end **508**-*d*. As a result, the expansion busbar **320** forms into a spring at the distal end **508**-*d* such that the spring is compressed and the two expansion sub-busbars **505** press against the panelboard busbars **306** when the expansion busbar **320** is received in the gap **332**, thereby increasing the coupling strength between the expansion subassembly **302** and the other bus subassembly **206**. In some embodiments, the expansion subassembly **302** may include one or more clips **402** coupled to one or more ends of the expansion busbars **320**.

[0048] In the example embodiment, because the expansion busbars **320** are exposed when the panelboard bus assembly **205** is not assembled with another bus subassembly **206**, an insulation cap **502** is placed on the expansion busbars **320** to provide protection against energized expansion busbars **320**. The insulation cap **502** is fabricated with electrically non-conductive material such as rubber and/or ceramic. The insulation cap **502** may include a handle **506** configured to facilitate the coupling of the insulation cap **502** with the panelboard bus assembly **205** or removal of the insulation cap **502** from the panelboard bus assembly **205**.

[0049] The expansion of the panelboard bus assembly **205** using the expansion assembly **302-5** saves footprint of the expanded panelboard bus assembly **205**, compared to the expanded panelboard bus assembly **205** using a distinct expansion subassembly **302**. To expand the panelboard bus assembly, a separate expansion subassembly **302** is not needed. Instead, another bus subassembly **206** or another panelboard bus assembly **205** is coupled with the panelboard bus assembly **205** via the expansion subassembly **302-5**.

[0050] FIG. **6** is a flow chart of an example method **600** of assembling an expandable electrical panelboard system. Electrical panelboard systems may be any panelboard system **200** described herein. In the example embodiments, the method **600** includes forming **602** an enclosure. The method **600** also includes forming **604** a panelboard bus assembly. Panelboard bus assemblies may be any panelboard bus assembly **205** described herein. The method **600** further includes positioning **606** the panelboard bus assembly in the enclosure. In some embodiments, the panelboard system **200** does not include an enclosure. The method **600** does not include forming **602** an enclosure or positioning **606** the panelboard bus assembly in the enclosure. In other embodiments, the panelboard bus assembly **205** may be assembled into an existing enclosure **102** of the panelboard system **200**. The method **600** does not include forming **602** an enclosure.

[0051] At least one technical effect of the systems and methods described herein includes (a) expandable panelboard bus assemblies; (b) an expansion subassembly for expanding a bus subassembly; (c) an expansion subassembly and a bus subassembly forming into one unit to reduce the footprint of the panelboard bus assembly; and (d) an expansion subassembly including mechanisms to increase the strength in the mechanical and electrical coupling of the expandable panelboard bus assembly.

[0052] Example embodiments of systems, assemblies, and methods of panelboards and panelboard buses are described above in detail. The systems and methods are not limited to the specific embodiments described herein but, rather, components of the systems and/or operations of the methods may be utilized independently and separately from other components and/or operations described herein. Further, the described components and/or operations may also be defined in, or used in combination with, other systems, methods, and/or devices, and are not limited to practice with only the systems described herein.

[0053] Although specific features of various embodiments of the invention may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the invention, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

[0054] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

What is claimed is:

1. An expandable electrical panelboard system, comprising:

a panelboard bus assembly, the panelboard bus assembly comprising:

a bus subassembly comprising:

one or more busbar insulators; and

one or more panelboard busbars at least partially enclosed in the one or more busbar insulators; and

an expansion subassembly comprising:

one or more expansion busbars, wherein the one or more expansion busbars are electrically conductive and at least partially exposed,

wherein the bus subassembly is configured to mechanically and electrically couple with another bus subassembly via the expansion subassembly.

2. The electrical panelboard system of claim **1**, wherein the bus subassembly is a distinct component from the expansion subassembly, the expansion subassembly further comprising:

one or more expansion insulators,

wherein the one or more expansion busbars are at least partially enclosed in the one or more expansion insulators.

3. The electrical panelboard system of claim **2**, wherein a side shape of the expansion subassembly is complementary with a side shape of the bus subassembly.

4. The electrical panelboard system of claim **1**, wherein the bus subassembly and the expansion subassembly are formed as one unit, the one or more expansion busbars extending from the one or more panelboard busbars and out of the one or more busbar insulators.

5. The electrical panelboard system of claim **4**, wherein the expansion busbar comprises a distal end and a proximal end opposite the distal end and proximal to the panelboard busbars, a thickness of the expansion busbar at the distal end being greater than a thickness of the expansion busbar at the proximal end.

6. The electrical panelboard system of claim **4**, wherein the panelboard bus assembly further comprises an insulation cap sized to receive the one or more expansion busbars.

7. The electrical panelboard system of claim **1**, wherein the one or more expansion busbars further include one or more clips positioned at one or more ends of the one or more expansion busbars.

8. An expandable panelboard bus assembly for an electrical panelboard system, the panelboard bus assembly comprising:

a bus subassembly comprising:
one or more busbar insulators; and
one or more panelboard busbars at least partially enclosed in the one or more busbar insulators; and
an expansion subassembly comprising:
one or more expansion busbars, wherein the one or more expansion busbars are electrically conductive and at least partially exposed,
wherein the bus subassembly is configured to mechanically and electrically couple with another bus subassembly via the expansion subassembly.

9. The panelboard bus assembly of claim **8**, wherein the bus subassembly is a distinct component from the expansion subassembly, the expansion subassembly further comprising:
one or more expansion insulators,
wherein the one or more expansion busbars are at least partially enclosed in the one or more expansion insulators.

10. The panelboard bus assembly of claim **9**, wherein a side shape of the expansion subassembly is complementary with a side shape of the bus subassembly.

11. The panelboard bus assembly of claim **9**, wherein the expansion subassembly further comprises an expansion bolt coupled with the one or more expansion insulators and the one or more expansion busbars.

12. The panelboard bus assembly of claim **8**, wherein the bus subassembly and the expansion subassembly are formed as one unit, the one or more expansion busbars extending from the one or more panelboard busbars and out of the one or more busbar insulators.

13. The panelboard bus assembly of claim **12**, wherein the expansion busbar comprises a distal end and a proximal end opposite the distal end and proximal to the panelboard busbars, a thickness of the expansion busbar at the distal end being greater than a thickness of the expansion busbar at the proximal end.

14. The panelboard bus assembly of claim **12**, further comprising an insulation cap sized to receive the one or more expansion busbars.

15. The panelboard bus assembly of claim **12**, wherein the panelboard busbar comprises a first panelboard sub-busbar and a second panelboard sub-busbar, the expansion busbar comprising a first expansion sub-busbar extending from the first panelboard sub-busbar and a second expansion sub-busbar extending from the second panelboard sub-busbar.

16. The panelboard bus assembly of claim **8**, wherein the expansion subassembly is interwoven with the bus subassembly.

17. The panelboard bus assembly of claim **8**, wherein the one or more expansion busbars further include one or more clips positioned at one or more ends of the one or more expansion busbars.

18. A method of assembling an expandable electrical panelboard system, the method including:
forming a panelboard bus assembly, the panelboard bus assembly including:
a bus subassembly including:
one or more busbar insulators; and
one or more panelboard busbars at least partially enclosed in the one or more busbar insulators; and
an expansion subassembly including:
one or more expansion busbars, wherein the one or more expansion busbars are electrically conductive and at least partially exposed, wherein the bus subassembly is configured to mechanically and electrically couple with another bus subassembly via the expansion subassembly.

19. The method of claim **18**, wherein the bus subassembly is a distinct component from the expansion subassembly.

20. The method of claim **18**, wherein forming the panelboard bus assembly further comprises forming the expansion subassembly and the bus subassembly as one unit.

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